

# Networking, Reconnaissance & Enumeration

Task 3 - NovaShyld Technologies Cybersecurity Internship

## 1. Passive Reconnaissance (OSINT)

To demonstrate passive reconnaissance capabilities, Open-Source Intelligence (OSINT) techniques were applied to a public target (facebook.com). This phase involves mapping the external attack surface without directly interacting with the target's internal network.

### 1.1. WHOIS & Domain Information

Initial reconnaissance was performed using the `whois` utility to extract domain registration details, registrar information, and authoritative name servers.

### 1.2. DNS Enumeration (nslookup & dig)

DNS records were queried to identify IP addresses, mail servers, and text records associated with the domain.

### 1.3. Subdomain Enumeration & Surface Mapping

Public databases and web-based OSINT tools were utilized to discover subdomains, open ports, and infrastructure mapping.

## 2. Active Network Scanning & Target Profiles

Active reconnaissance was performed using **Nmap** to map out the isolated VAPT lab environment, perform OS fingerprinting, and enumerate running services and their specific versions.

Target 1: Vulnerable Web Application (DVWA)

**IP Address:** 192.168.100.130

**MAC Address:** 00:0C:29:C0:F8:F4 (VMware)

**Operating System:** Ubuntu Linux

**Network Distance:** 1 hop

| Port | Protocol | State | Service | Version / Banner                 |
|------|----------|-------|---------|----------------------------------|
| 22   | TCP      | Open  | ssh     | OpenSSH 10.2p1 Ubuntu 2ubuntu3.5 |
| 8081 | TCP      | Open  | http    | Apache httpd 2.4.25 ((Debian))   |

Target 2: Metasploitable2 (Linux)

**IP Address:** 192.168.100.132

**MAC Address:** 00:0C:29:8C:25:C1 (VMware)

**Operating System:** Linux 2.6.9 - 2.6.33

**Network Distance:** 1 hop

| Port | Protocol | State | Service     | Version / Banner                    |
|------|----------|-------|-------------|-------------------------------------|
| 21   | TCP      | Open  | ftp         | vsftpd 2.3.4                        |
| 22   | TCP      | Open  | ssh         | OpenSSH 4.7p1 Debian 8ubuntu1       |
| 23   | TCP      | Open  | telnet      | Linux telnetd                       |
| 25   | TCP      | Open  | smtp        | Postfix smtpd                       |
| 53   | TCP      | Open  | domain      | ISC BIND 9.4.2                      |
| 80   | TCP      | Open  | http        | Apache httpd 2.2.8 ((Ubuntu) DAV/2) |
| 111  | TCP      | Open  | rpcbind     | 2 (RPC #100000)                     |
| 139  | TCP      | Open  | netbios-ssn | Samba smbd 3.X - 4.X                |
| 445  | TCP      | Open  | netbios-ssn | Samba smbd 3.X - 4.X                |
| 512  | TCP      | Open  | exec        | netkit-rsh rexecd                   |

| Port | Protocol | State | Service    | Version / Banner                    |
|------|----------|-------|------------|-------------------------------------|
| 513  | TCP      | Open  | login      | OpenBSD or Solaris rlogind          |
| 514  | TCP      | Open  | tcpwrapped | -                                   |
| 1099 | TCP      | Open  | java-rmi   | GNU Classpath gmiregistry           |
| 1524 | TCP      | Open  | bindshell  | Metasploitable root shell           |
| 2049 | TCP      | Open  | nfs        | 2-4 (RPC #100003)                   |
| 2121 | TCP      | Open  | ftp        | ProFTPD 1.3.1                       |
| 3306 | TCP      | Open  | mysql      | MySQL 5.0.51a-3ubuntu5              |
| 3632 | TCP      | Open  | distccd    | distccd v1                          |
| 5432 | TCP      | Open  | postgresql | PostgreSQL DB 8.3.0 - 8.3.7         |
| 5900 | TCP      | Open  | vnc        | VNC (protocol 3.3)                  |
| 6000 | TCP      | Open  | X11        | (access denied)                     |
| 6667 | TCP      | Open  | irc        | UnrealIRCd                          |
| 6697 | TCP      | Open  | irc        | UnrealIRCd                          |
| 8009 | TCP      | Open  | ajp13      | Apache Jserv (Protocol v1.3)        |
| 8180 | TCP      | Open  | http       | Apache Tomcat/Coyote JSP engine 1.1 |
| 8787 | TCP      | Open  | drb        | Ruby DRb RMI                        |